ROUND REINFORCED CONCRETE PIPE APPROXIMATE WEIGHT PER FOOT

INSIDE DIAMETER		WEIGHT PER FT.	
mm	ft/in	kg	lb
305	**12"**	42	**93**
380	**15"**	58	**127**
460	**18"**	76	**168**
530	**1'9"**	97	**214**
610	**2'0"**	120	**265**
685	**2'3"**	146	**322**
760	**2'6"**	174	**384**
840	**2'9"**	205	**452**
915	**3'0"**	238	**524**
1070	**3'6"**	311	**686**
1220	**4'0"**	393	**867**
1370	**4'6"**	485	**1069**
1525	**5'0"**	588	**1295**
1675	**5'6"**	699	**1542**
1830	**6'0"**	821	**1811**
1980	**6'6"**	952	**2100**
2135	**7'0"**	1093	**2409**
2285	**7'6"**	1242	**2740**
2440	**8'0"**	1402	**3090**
2590	**8'6"**	1578	**3480**
2740	**9'0"**	1753	**3865**

NOTE: Table courtesy of American Concrete Pipe Assn.

COEFFICIENT OF TRACTION FACTORS

MATERIAL	TRACTION FACTORS	
	Rubber Tires	Tracks
Concrete	0.90	0.45
Clay loam, dry	0.55	0.90
Clay loam, wet	0.45	0.70
Rutted clay loam	0.40	0.70
Dry sand	0.20	0.30
Wet sand	0.40	0.50
Quarry pit	0.65	0.55
Gravel road (loose not hard)	0.36	0.50
Packed snow	0.20	0.27
Ice	0.12	0.12
Semi-skeleton shoes		
Firm earth	0.55	0.90
Loose earth	0.45	0.60
Coal, stockpiled	0.45	0.60

NOTE: The elevated sprocket design Track-Type Tractors (D11, D10T2, D9T, D9R, D8T, and D8R), with their suspended undercarriage, provide up to 15% more efficient tractive effort than rigid tracked Track-Type Tractors.

SPEED CONVERSION

km/h Equivalents in m/min				MPH Equivalents in FPM			
km/h	m/min	km/h	m/min	mph	fpm	mph	fpm
1	16.7	21	350.0	**1**	**88**	**21**	**1848**
2	33.3	22	366.7	**2**	**176**	**22**	**1936**
3	50.0	23	383.3	**3**	**264**	**23**	**2024**
4	66.7	24	400.0	**4**	**352**	**24**	**2112**
5	83.3	25	416.7	**5**	**440**	**25**	**2200**
6	100.0	26	433.3	**6**	**528**	**26**	**2288**
7	116.7	27	450.0	**7**	**616**	**27**	**2376**
8	133.3	28	466.7	**8**	**704**	**28**	**2464**
9	150.0	29	483.3	**9**	**792**	**29**	**2552**
10	166.7	30	500.0	**10**	**880**	**30**	**2640**
11	183.3	31	516.7	**11**	**968**	**31**	**2728**
12	200.0	32	533.3	**12**	**1056**	**32**	**2816**
13	216.7	33	550.0	**13**	**1144**	**33**	**2904**
14	233.3	34	566.7	**14**	**1232**	**34**	**2992**
15	250.0	35	583.3	**15**	**1320**	**35**	**3080**
16	266.7	36	600.0	**16**	**1408**	**36**	**3168**
17	283.3	37	616.7	**17**	**1496**	**37**	**3256**
18	300.0	38	633.3	**18**	**1584**	**38**	**3344**
19	316.7	39	650.0	**19**	**1672**	**39**	**3432**
20	333.3	40	666.7	**20**	**1760**	**40**	**3520**

NOTE: Since 1 km/h equals 16.7 m/min (1000 ÷ 60), to interpolate add 1.67 m/min for each 0.1 km/h.

NOTE: Since 1 mph equals 88 fpm (5280 ÷ 60), to interpolate add 8.8 fpm for every 0.1 mph.

1 mph = 26.9 m/min.

BEARING POWERS

MATERIAL	BEARING POWER Bar	lb/in²	Metric t/m²	U.S. tons/ft²
Rock (semi- shattered)	4.8	**70**	50	**5**
Rock (solid)	24.1	**350**	240	**24**
Clay, dry.	3.8	**55**	40	**4**
medium dry	1.9	**27**	20	**2**
soft	1.0	**14**	10	**1**
Gravel, cemented	7.6	**110**	80	**8**
Sand, compact dry	3.8	**55**	40	**4**
clean dry	1.9	**27**	20	**2**
Quicksand & alluvial soil . .	0.5	**7**	5	**0.5**

AGRICULTURAL COMMODITIES CONVERSION FACTORS

	lb	kg	Metric Ton
1 Bushel of Corn*	56	25.40	0.02540
1 Bushel of Soybean*	60	27.22	0.02721
1 Bushel of Oats*	32	14.51	0.01451
1 Bushel of Wheat*	60	27.22	0.02721
1 Bale of Cotton	478	216.81	0.21681

1 metric ton of Corn	39.37 Bushels*
1 metric ton of Soybean	36.75 Bushels*
1 metric ton of Oats	68.92 Bushels*
1 metric ton of Wheat	36.75 Bushels*
1 metric ton of Cotton	4.61 Bales

*Bushel is a volume measurement, 1 Bushel = 35.24 liters = 9.31 U.S. Gallons. In the agricultural mercantile exchange, the Bushel is widely used for grains as weight. For the above weights, the market assumes a standard density for each type of grain.

CURVE SUPERELEVATION IN PERCENT GRADE, TO PROVIDE NO LATERAL TIRE FORCE

Negotiating curves can generate high lateral tire forces. These forces contribute to high tire wear and ply separation. Superelevating the curve helps eliminate these forces. The amount of superelevation depends on the curve's radius and the speed at which it is negotiated.

The following table is a guide for providing the superelevation necessary to eliminate lateral forces.

Superelevated turns present a danger when slippery. For this reason, curves superelevated over 10% should be used with caution. Unless the proper speed is maintained, matching the elevation of the curve, a vehicle may slide off of the lower edge of the roadway. Superelevated curves should be maintained in good tractive conditions.

TURN RADIUS m	ft	Speed 16 km/h 10 mph	Speed 24 km/h 15 mph	Speed 32 km/h 20 mph	Speed 40 km/h 25 mph	Speed 48 km/h 30 mph	Speed 56 km/h 35 mph	Speed 64 km/h 40 mph	Speed 72 km/h 45 mph
15.2	**50**	13%	30%	—	—	—	—	—	—
30.5	**100**	7%	15%	27%	—	—	—	—	—
45.7	**150**	4%	10%	18%	28%	—	—	—	—
61.0	**200**	3%	8%	13%	21%	30%	—	—	—
91.5	**300**	2%	5%	9%	14%	20%	27%	—	—
152.4	**500**	1%	3%	5%	8%	12%	16%	21%	27%
213.4	**700**	1%	2%	4%	6%	9%	12%	15%	19%
304.9	**1000**	1%	2%	3%	4%	6%	8%	11%	14%

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MAXIMUM SPEED ON CURVES FOR VARIOUS SUPERELEVATION GRADES WITH A 0.20 LATERAL COEFFICIENT OF TRACTION

Another approach to superelevated curves is to determine the safe speed for negotiating a turn at a certain lateral tire force. In general, a 20% lateral coefficient of traction is conservative for all but ice and slippery conditions, making table values safe to use for most applications. The following table shows maximum speed with various superelevations to maintain a 0.20 lateral coefficient of traction.

A transition section may be necessary at higher speeds when entering or departing from a superelevated turn.

TURN RADIUS m	ft	Flat Curve km/h	mph	5% Superelevation km/h	mph	10% Superelevation km/h	mph
7.6	**25**	14	**9**	16	**10**	17	**11**
15.2	**50**	20	**12**	22	**14**	24	**15**
30.5	**100**	28	**17**	31	**19**	34	**21**
45.7	**150**	34	**21**	38	**24**	42	**26**
61.0	**200**	39	**24**	44	**27**	48	**30**
91.5	**300**	48	**30**	54	**34**	59	**37**
152	**500**	62	**39**	70	**43**	76	**47**
213	**700**	74	**46**	—	—	—	—

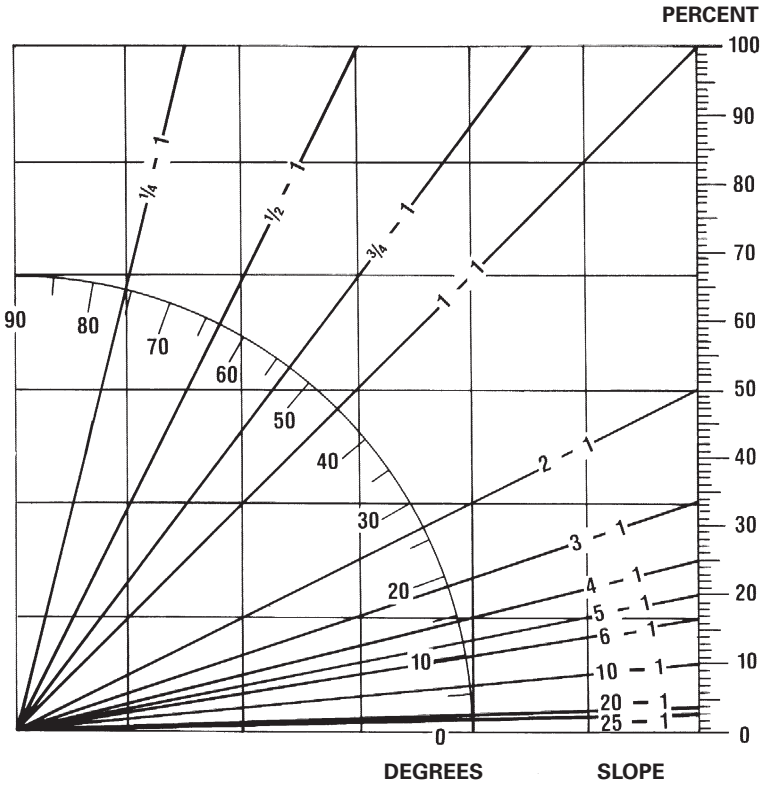
WEIGHT* OF MATERIALS	LOOSE		BANK		LOAD FACTORS
	kg/m³	lb/yd³	kg/m³	lb/yd³	
Basalt	1960	**3300**	2970	**5000**	0.67
Bauxite, Kaolin	1420	**2400**	1900	**3200**	0.75
Caliche	1250	**2100**	2260	**3800**	0.55
Carnotite, uranium ore	1630	**2750**	2200	**3700**	0.74
Cinders	560	**950**	860	**1450**	0.66
Clay — Natural bed	1660	**2800**	2020	**3400**	0.82
Dry	1480	**2500**	1840	**3100**	0.81
Wet	1660	**2800**	2080	**3500**	0.80
Clay & gravel — Dry	1420	**2400**	1660	**2800**	0.85
Wet	1540	**2600**	1840	**3100**	0.85
Coal — Anthracite, Raw	1190	**2000**	1600	**2700**	0.74
Washed	1100	**1850**			0.74
Ash, Bituminous Coal	530-650	**900-1100**	590-890	**1000-1500**	0.93
Bituminous, Raw	950	**1600**	1280	**2150**	0.74
Washed	830	**1400**			0.74
Decomposed rock —					
75% Rock, 25% Earth	1960	**3300**	2790	**4700**	0.70
50% Rock, 50% Earth	1720	**2900**	2280	**3850**	0.75
25% Rock, 75% Earth	1570	**2650**	1960	**3300**	0.80
Earth — Dry packed	1510	**2550**	1900	**3200**	0.80
Wet excavated	1600	**2700**	2020	**3400**	0.79
Loam	1250	**2100**	1540	**2600**	0.81
Granite — Broken	1660	**2800**	2730	**4600**	0.61
Gravel — Pitrun	1930	**3250**	2170	**3650**	0.89
Dry	1510	**2550**	1690	**2850**	0.89
Dry 6-50 mm (1/4"-2")	1690	**2850**	1900	**3200**	0.89
Wet 6-50 mm (1/4"-2")	2020	**3400**	2260	**3800**	0.89
Gypsum — Broken	1810	**3050**	3170	**5350**	0.57
Crushed	1600	**2700**	2790	**4700**	0.57
Hematite, iron ore, high grade	1810-2450	**4000-5400**	2130-2900	**4700-6400**	0.85
Limestone — Broken	1540	**2600**	2610	**4400**	0.59
Crushed	1540	**2600**	—	**—**	—
Magnetite, iron ore	2790	**4700**	3260	**5500**	0.85
Pyrite, iron ore	2580	**4350**	3030	**5100**	0.85
Sand — Dry, loose	1420	**2400**	1600	**2700**	0.89
Damp	1690	**2850**	1900	**3200**	0.89
Wet	1840	**3100**	2080	**3500**	0.89
Sand & clay — Loose	1600	**2700**	2020	**3400**	0.79
Compacted	2400	**4050**			
Sand & gravel — Dry	1720	**2900**	1930	**3250**	0.89
Wet	2020	**3400**	2230	**3750**	0.91
Sandstone	1510	**2550**	2520	**4250**	0.60
Shale	1250	**2100**	1660	**2800**	0.75
Slag — Broken	1750	**2950**	2940	**4950**	0.60
Snow — Dry	130	**220**			
Wet	520	**860**			
Stone — Crushed	1600	**2700**	2670	**4500**	0.60
Taconite	1630-1900	**3600-4200**	2360-2700	**5200-6100**	0.58
Top Soil	950	**1600**	1370	**2300**	0.70
Taprock — Broken	1750	**2950**	2610	**4400**	0.67
Wood Chips**	—	**—**	—	**—**	—

*Varies with moisture content, grain size, degree of compaction, etc. Tests must be made to determine exact material characteristics.

**Weights of commercially important wood species can be found in the last pages of the Logging & Forest Products section. To obtain wood weights use the following equations: $lb/yd^3 = (lb/ft^3) \times .4 \times 27$

$kg/m^3 = (kg/m^3) \times .4$

GRADE COMPARISON CHART
DEGREES — PERCENT — SLOPE



GRADE IN DEGREES AND PERCENTS

DEGREES	PERCENT
1	1.8
2	3.5
3	5.2
4	7.0
5	8.8
6	10.5
7	12.3
8	14.0
9	15.8
10	17.6
11	19.4
12	21.3
13	23.1
14	24.9
15	26.8
16	28.7
17	30.6
18	32.5
19	34.4
20	36.4
21	38.4
22	40.4
23	42.4
24	44.5
25	46.6
26	48.8
27	51.0
28	53.2
29	55.4
30	57.7
31	60.0
32	62.5
33	64.9
34	67.4
35	70.0
36	72.7
37	75.4
38	78.1
39	81.0
40	83.9
41	86.9
42	90.0
43	93.3
44	96.6
45	100.0

CONVERSION FACTORS

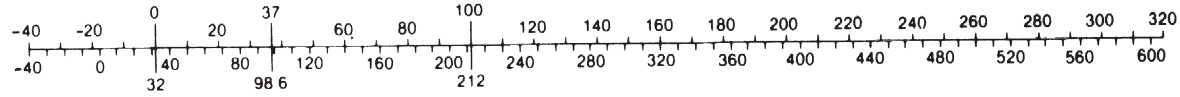
Multiply Metric Unit	By	To Obtain English Unit
kilometer (km)	0.6214	mile
meter (m)	1.0936	yard
meter (m)	3.28	foot
centimeter (cm)	0.0328	foot
millimeter (mm)	0.03937	inch
sq kilometer (km^2)	0.3861	square mile
hectare (ha)	2.471	acre
sq meter (m^2)	10.764	square foot
sq meter (m^2)	1550	square inch
sq centimeter (cm^2)	0.1550	square inch
cu centimeter (cm^3)	0.061	cubic inch
cu meter (m^3)	1.308	cubic yard
liter (L)	61.02	cubic inch
liter (L)	0.001308	cubic yard
km/h	0.621	mph
liter (L)	0.2642	U.S. gallon
liter (L)	0.22	Imperial gallon
metric ton (t)	0.984	long ton
metric ton (t)	1.102	short ton
kilogram (kg)	2.205	pound, avdp.
gram (g or gr)	.0353	ounce, avdp.
kilonewton (kN)	225	pound (force)
newton (N)	0.225	pound (force)
cu centimeter (cm^3)	0.0338	fluid ounce
kilograms/cu meter	1.686	pounds/cu yd
kilograms/cu meter	0.062	pounds/cu ft
kilograms/sq cm (kg/cm^2)	14.225	pounds/sq in
kilocalorie (kcal)	3.968	Btu
kilogram-meter (kg•m)	7.233	foot-pound
meter-kilogram (m•kg)	7.233	pound-foot
metric horsepower (CV)	0.9863	hp
kilowatt (kW)	1.341	hp
kilopascal (kPa)	0.145	psi
bar	14.5	psi
tons/m^3	1692	pounds/cu yd
decaliter	0.283	bushel

Multiply English Unit	By	To Obtain Metric Unit
mile, statute (m)	1.609	kilometer
yard (yd)	0.9144	meter
foot (ft)	0.3048	meter
inch (in)	25.4	millimeter
sq mile ($mile^2$)	2.590	sq kilometer
acre	0.4047	hectare
sq foot (ft^2)	0.0929	sq meter
sq inch (in^2)	0.000645	sq meter
cu yard (yd^3)	0.7645	cu meter
cu inch (in^3)	16.387	cu centimeter
cu foot (ft^3)	0.0283	cu meter
cu inch (in^3)	0.0164	liter
cubic yard (yd^3)	764.55	liter
mph	1.61	km/h
Ton — mph	1.459	tkm/h
U.S. gallon (US Gal)	3.785	liter
U.S. gallon	0.833	Imperial gallon
long ton (lg ton)	1.016	metric ton
short ton (sh ton)	0.907	metric ton
pound (lb)	0.4536	kilogram
ounce (oz)	28.35	gram
pound (lb) (force)	0.00445	kilonewton
pound (lb) (force)	4.45	newton
fluid oz (fl oz)	29.57	cu centimeter
lb/cu ft (lb/ft^3)	16.018	kg/cu meter
lb/cu yd (lb/yd^3)	0.5933	kg/cu meter
pounds/sq. in.	0.0703	kilogram/sq cm
psi	0.0689	bar
psi	6.89	kilopascal
Btu	0.2520	kilogram-calorie
foot-pound (ft-lb)	0.1383	kilogram-meter
horsepower (hp)	1.014	metric horsepower
horsepower (hp)	0.7457	kilowatt
pounds/cu yd	0.0005928	tons/m^3
pounds (No. 2 diesel fuel)	0.1413	U.S. gallon
bushel	3.524	decaliter

NOTE: Some of the above factors have been rounded for convenience. For exact conversion factors please consult International System of Units (SI) table.

Temperature conversion

Degree C



Degree F

°C = (°F – 32) ÷ 1.8

°F = (C × 1.8) + 32

METRIC UNIT EQUIVALENTS

1 km	=	1000 m
1 m	=	100 cm
1 cm	=	10 mm
1 km^2	=	100 ha
1 ha	=	10,000 m^2
1 m^2	=	10,000 cm^2
1 cm^2	=	100 mm^2
1 m^3	=	1000 liters
1 liter	=	1000 cm^3
1 metric ton	=	1000 kg
1 quintal	=	100 kg
1 N	=	0.10197 kg•m/s^2
1 kg	=	1000 g
1 g	=	1000 mg
1 bar	=	14.504 psi
1 cal	=	427 kg•m
	=	0.0016 cv•h
	=	0.00116 kw•h
torque unit		
1 CV	=	75 kg•m/s
1 kg/cm^2	=	0.97 atmosph.

ENGLISH UNIT EQUIVALENTS

1 mile	=	1760 yd
1 yd	=	3 ft
1 ft	=	12 in
1 sq mile	=	640 acres
1 acre	=	43,560 sq ft
1 sq ft	=	144 sq in
1 cu ft	=	7.48 gal liq
1 gal	=	231 cu in
	=	4 quarts liq
1 quart	=	32 fl oz
1 fl oz	=	1.80 cu in
1 sh ton	=	2000 lb
1 lg ton	=	2240 lb
1 lb	=	16 oz, avdp
1 Btu	=	778 ft lb
	=	0.000393 hph
	=	0.000293 kwh
1 mechanical hp	=	550 ft-lb/sec
1 atmosph.	=	14.7 lb/in^2

POWER UNIT EQUIVALENTS

kW	=	Kilowatt
hp	=	Mechanical Horsepower
CV	=	Cheval Vapeur (Steam Horsepower)
		French Designation For **Metric** Horsepower
PS	=	Pferdestärke (Horsepower)
		German Designation For **Metric** Horsepower
1 hp	=	1.014 CV = 1.014 PS
	=	0.7457 kW
1 PS	=	1 CV = 0.986 hp
	=	0.7355 kW
1 kW	=	1.341 hp
	=	1.36 CV
	=	1.36 PS

MACHINE/ENGINE CROSS REFERENCE

NOTE: This list stopped being updated since PHB49.

Machine Model	Engine Model
Track-Type Tractors	
D3K XL/LGP	C4.4 ACERT™
D4K XL/LGP	C4.4 ACERT
D5K XL/LGP	C4.4 ACERT
D5R	C6.6 ACERT
D5R2 XL/LGP	C7.1 ACERT
D5T XL	C7.1 ACERT
D6K2 XL/LGP[2]	C7.1 ACERT
D6K2 XL/LGP	C4.4 ACERT
D6N XL/LGP	C6.6 ACERT
D6R2	C9 ACERT
D6T	C9 ACERT
D6T[1]	C9.3 ACERT
D7E	C9.3 ACERT
D7R	C9 ACERT
D8R/D8R LGP	3406C TA
D8T/D8T LGP	C15 ACERT
D9R	3408C TA
D9T	C18 ACERT
D10T2	C27 ACERT
D11/D11 CD	C32 ACERT
Motor Graders	
120K	C7 ACERT
120K2	C7 ACERT
120M	C6.6 ACERT
120M AWD	C6.6 ACERT
120M2	C7.1 ACERT
120M2 AWD	C7.1 ACERT
12K	C7 ACERT
12M	C7 ACERT
12M2	C9.3 ACERT
12M2 AWD	C9.3 ACERT
12M3	C9.3 ACERT
12M3 AWD	C9.3 ACERT
140K	C7 ACERT
140K2	C7 ACERT
140M	C7 ACERT
140M AWD	C9 ACERT
140M2	C9.3 ACERT
140M2 AWD	C9.3 ACERT
140M3	C9.3 ACERT
140M3 AWD	C9.3 ACERT
160K	C7 ACERT
160M	C9 ACERT
160M AWD	C9 ACERT
160M2	C9.3 ACERT
160M2 AWD	C9.3 ACERT
160M3	C9.3 ACERT
160M3 AWD	C9.3 ACERT
14M3	C13 ACERT
16M3	C13 ACERT
18M3	C13 ACERT
24M	C18 ACERT

Machine Model	Engine Model
Excavators	
311F L	C3.4B
312D2/312D2 L	3054C
312D2 GC	3054C
312E	C4.4 ACERT
313D2	3054C
313D2/313D2 L	C4.4 ACERT
313D2 GC	C4.4 ACERT
313F L	C4.4 ACERT
313F L GC	C4.4 ACERT
314E/314E L	C4.4 ACERT
315F L	C4.4 ACERT
316E L	C4.4 ACERT
316F L	C4.4 ACERT
318D2 L	3054C
318D2 L	C4.4 ACERT
318F L	C4.4 ACERT
M313D	C4.4 ACERT
M315D	C4.4 ACERT
M315D2	C4.4 ACERT
M316D	C6.6 ACERT
M317D2	C4.4 ACERT
M318D	C6.6 ACERT
M318F	C7.1 ACERT
M320D2	C7.1
M320F	C7.1 ACERT
M322D	C6.6 ACERT
320D	C7.1 ACERT
320D2 GC	C4.4 ACERT
323D2	C7.1 ACERT
320E	C6.6 ACERT
320E RR	C6.6 ACERT
320F	C4.4 ACERT
323F	C7.1 ACERT
323F N	C4.4 ACERT
325F	C4.4 ACERT
326D2	C7.1 ACERT
326F	C7.1 ACERT
330D2	C7.1 ACERT
330F	C7.1 ACERT
335F	C7.1 ACERT
336F/336F XE	C9.3 ACERT
336D2/340D2	C9 ACERT
336D2 XE	C9 ACERT
349D2/349D2 L	C13 ACERT
349F L/352F	C13 ACERT
349F L XE/352F XE	C13 ACERT
374F L	C15 ACERT
390F L	C18 ACERT
5110B	3412B HEUI™
5130B	3508B (EUI) TA
5230B	3516B (EUI) TA

Machine Model	Engine Model
Pipelayers	
PL61	C6.6 ACERT
PL72	C9.3 ACERT
PL83	C15 ACERT
PL87	C15 ACERT
Wheel Tractor-Scrapers	
621K, 623K, 627K Tractor	C13 ACERT
627K Scraper	C9 ACERT
631K	C18 ACERT
637K Tractor	C18 ACERT
637K Scraper	C9 ACERT
657G Tractor	C18 ACERT
657G Scraper	C15 ACERT
Forest Machines	
320D Series 2 FM[2]	C7.1 ACERT
320D Series 2 FM LL[2]	C7.1 ACERT
538[1]	C7.1 ACERT
538 LL[1]	C7.1 ACERT
324D FM[2]	C7 ACERT
324D FM LL[2]	C7 ACERT
325D FM[2]	C7 ACERT
325D FM LL[2]	C7 ACERT
558 LL[1]	C7.1 ACERT
568[3]	C9.3 ACERT
568 LL[3]	C9.3 ACERT
Track Harvesters	
501HD[2]	C6.6 ACERT
521B[2]	C9 ACERT
522B[2]	C9 ACERT
541 Series 2[2]	C9 ACERT
552 Series 2[2]	C9 ACERT
Wheel Harvester	
550	C7 ACERT
Wheel Skidders	
525D	C7.1 ACERT
535D	C7.1 ACERT
545D	C7.1 ACERT
555D	C7.1 ACERT
Track Skidders	
517	3304 TA
527	3304 TA

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
[2] Emits equivalent to Tier 3/Stage IIIA/Japan 2006 (Tier 3).
[3] Emits equivalent to Tier 4 Interim/Stage IIIB/Japan 2011 (Tier 4 Interim).

MACHINE/ENGINE CROSS REFERENCE (Continued)

Machine Model	Engine Model
Forwarders	
564	C6.6 ACERT
574	C6.6 ACERT
584	C7 ACERT
584HD	C7 ACERT
Track Feller Bunchers	
521B[2]	C9 ACERT
522B[2]	C9 ACERT
541 Series 2[2]	C9 ACERT
552 Series 2[2]	C9 ACERT
Knuckleboom Loaders	
519[2]	C6.6
529[2]	C6.6
559C	C6.6 ACERT
569[2]	C6.6
579C	C6.6 ACERT
Wheel Feller Bunchers	
553	C6.6 ACERT
563	C7 ACERT
573	C7 ACERT
Mining & Off-Highway Trucks	
770G[1]	C15 ACERT
770G	C15 ACERT
772G[1]	C18 ACERT
772G	C18 ACERT
773G[1]	C27 ACERT
773G	C27 ACERT
775G[1]	C27 ACERT
775G	C27 ACERT
777D	3508B (EUI) TA
777G[1]	C32 ACERT
777G	C32 ACERT
785C	3512B (EUI) TA
785D	3512C HD (EUI) ATAAC
789C	3516B (EUI) TA
793D	3516B (EUI) TA
793F	C175-16 (EUI) ATAAC
797F	C175-20 (EUI) ATAAC

Machine Model	Engine Model
Articulated Trucks	
725	C11 ATAAC
730	C11 ATAAC
730 Ejector	C11 ATAAC
735	C15 ATAAC
740	C15 ATAAC
740 Ejector	C15 ATAAC
Wheel Dozers	
814F[3]	3176C ATAAC
814F II[2]	C9 ACERT
824H[2]	C15 ACERT
824K[1]	C15 ACERT
834K	C18 ACERT
844K	C27 ACERT
854K	C32 ACERT
Soil Compactors	
815F II[2]	C9 ACERT
815K[1]	C7.1 ACERT
825H[2]	C15 ACERT
825K[1]	C15 ACERT
Landfill Compactors	
816F	3176 TA
816F II	C9 ACERT
826G Series II	3406E TA
826H	C15 ACERT
836G	3456 TA
836H	C18 ACERT

Machine Model	Engine Model
Wheel Loaders/Integrated Toolcarriers	
950 GC	C7.1 ACERT
950H	C7 ACERT
950K	C7.1 ACERT
950L	C7.1 ACERT
950M	C7.1 ACERT
962H	C7 ACERT
962K	C7.1 ACERT
962L	C7.1 ACERT
962M	C7.1 ACERT
966H	C11 ACERT
966K	C9.3 ACERT
966L	C9.3 ACERT
966M	C9.3 ACERT
966M XE	C9.3 ACERT
972H	C13 ACERT
972K	C9.3 ACERT
972L	C9.3 ACERT
972M	C9.3 ACERT
972M XE	C9.3 ACERT
980H	C15 ACERT
980K	C13 ACERT
980L	C13 ACERT
980M	C13 ACERT
982M	C13 ACERT
986H	C15 ACERT
988K	C18 ACERT
990K	C27 ACERT
992K	C32 ACERT
993K	C32 ACERT
994K	3516E HD MUI
Track Loaders	
953D	C6.6 ACERT
953K	C7.1 ACERT
963D	C6.6 ACERT
963K	C7.1 ACERT
973C	C9 ATAAC

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
[2] Emits equivalent to Tier 3/Stage IIIA/Japan 2006 (Tier 3).
[3] Emits equivalent to Tier 2/Stage II/Japan 2001 (Tier 2).
NOTE: All machines are not available in all regions. Contact your local Cat dealer for product availability.

MACHINE/ENGINE CROSS REFERENCE (Continued)

Machine Model	Engine Model
Wheel Material Handlers	
M318D MH	**C6.6 ACERT**
M322D MH	C6.6 ACERT
M325D MH	C7 ACERT
M325D LMH	C7 ACERT
MH3037[2]	C7.2 ACERT
MH3037[1]	C7.1 ACERT
MH3049	C9 ACERT
MH3059	C9 ACERT
Track Material Handlers	
385C MH	C18 ACERT
Paving Products	
Cold Planers	
PM102	C7 ACERT
PM620	C18 ACERT
PM622	C18 ACERT
Reclaimer/Soil Stabilizers	
RM300	C11 ACERT
RM500B	C15 ACERT
Asphalt Pavers	
AP255E	C2.2
AP300F	C3.3B
AP355F	C3.3B
AP500F	C4.4 ACERT
AP555F	C4.4 ACERT
AP600F	C4.4 ACERT
	C7.1 ACERT
AP655F	C4.4 ACERT
	C7.1 ACERT
AP1000F	C7.1 ACERT
AP1055F	C7.1 ACERT

Machine Model	Engine Model
Paving Products (continued)	
Single Drum Vibratory Soil Compactors	
CS34	C3.4B
CP34	C3.4B
CS423E	3054C
CS44	C4.4 ACERT
CP44	C4.4 ACERT
CS44B	C3.4B
CP44B	C3.3B
CS533E	3054C
CS533E XT	3054C
CP533E	3054C
CS54B	C4.4B ACERT
CP54B	C4.4B ACERT
CS56B	C4.4B ACERT
	C6.6 ACERT
CP56B	C4.4B ACERT
	C6.6 ACERT
CS64B	C4.4B
CS66B	C4.4B
CS68B	C4.4B ACERT
	C6.6 ACERT
CP68B	C4.4B ACERT
	C6.6 ACERT
CS74B	C4.4B ACERT
	C6.6 ACERT
CP74B	C4.4B ACERT
	C6.6 ACERT
CS76B	C4.4B ACERT
CS78B	C4.4B ACERT
	C6.6 ACERT
CS79B	C6.6 ACERT

Machine Model	Engine Model
Paving Products (continued)	
Double Drum and Combi	
CB14B	KDW1003
CB22B	KDW1003
CB24B XT	C1.5
CC24B	C1.5
CB32B	C1.5
CB34B	C2.2
CC34B	C2.2
CB36B	C2.2
CB44B	C3.4B
	C4.4 ACERT
CD44B	C3.4
	C4.4 ACERT
CB46B	C3.4B
CB54B	C4.4 ACERT
CD54B	C3.4
CB64B	C4.4 ACERT
CB66B	C4.4 ACERT
CB68B	C4.4 ACERT
Pneumatic Tire Compactors	
CW14	C3.4B
CW16	C3.4B
	C4.4 ACERT
PS150C	3054C
CW34	C4.4 ACERT

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
[2] Emits equivalent to Tier 3/Stage IIIA/Japan 2006 (Tier 3).

ENGINE/MACHINE CROSS REFERENCE

Engine Model and (Cylinders)	Machine	Aspiration	Fuel Injection System	Bore × Stroke		Displacement	
				mm	in	L	in³
3013C (3)	CB-214E, CB-224E, CB-225E	NA	DI	75 × 72	**2.95 × 3.54**	1.50	**91.3**
C1.3	303E	NA	DI	78 × 88	**3.1 × 3.5**	1.3	**77**
C1.7	303.5E2 CR	NA	DI	87 × 92.4	**3.4 × 3.6**	1.7	**104**
C1.8	303.5E, 901C2 (Japan)	NA	DI	87 × 102.4	**3.4 × 4.0**	1.8	**111.4**
C2.2	CB-334E, CB-335E, 216B3	NA	DI	84 × 100	**3.31 × 3.94**	2.22	**135**
	226B3, 226D	T					
C2.4	304E2 CR, 304.5E2 XTC, 305E2 CR, 305.5E2 CR, 902C2 (Japan), 903C2	NA	DI	87 × 102.4	**3.4 × 4.0**	2.4	**146**
	307E2, 306E2	T					
C2.6	307E, 307E2 (China)	T	DI	87 × 110	**3.4 × 4.3**	2.6	**159.6**
C3.3B	308E2 CR, 308E2 VAB, 906K/M, 907K/M, 908K/M, 236D, 242D, 246D, 257D, 259D, 262D, 277D, 279D, 287D, 289D	T	DI	94 × 120	**3.7 × 4.7**	3.3	**203.3**
C3.8	272D2, 272D2 XHP, 297D2, 297D2 XHP, 299D2, 299D2 XHP	T	DI	100 × 120	**3.9 × 4.7**	3.8	**230**
Yanmar 3TNV70	300.9D	NA	IDI	70 × 74	**2.8 × 2.9**	0.854	**52.1**
Yanmar 3TNV76	301.4C, 301.7D, 301.7D CR, 302.2D, 302.4D, 302.7D	NA	IDI	76 × 82	**3.0 × 3.2**	1.12	**68.1**
3046 (I-6)	D3G XL, D3G LGP, D4G XL, CS-533E, D4G LGP, D5G XL, D5G LGP, CP-533E, 315C/315C L*, 939C	NA	DI	94 × 120	**3.7 × 4.7**	5.0	**305**
		T	DI				
3054C (I-4)	416E, 416F, 420F, 430F, 450F, 422F, 428F, 432F, 434F, 444F	NA/T***	DI	105 × 127	**4.13 × 5.0**	4.4	**268**
3054C	312D2/312D2 L, 313D2	T	DI	105 × 127	**4.13 × 5.0**	4.4	**268**
	318D2 L	TA					
3054E (I-4)	CB-434D, CS-323C, CS-423E, CP-323C, PS-150C	NA	DI	105 × 127	**4.13 × 5.0**	4.4	**268**
	AP-800C, BG-230, BG-650, 908, PS-360B, PF-300B, PS-300B, CS-433E, CP-433E, CB-534C	T (optional)					
	M313C, M315C, AP-650B, 315C L**, BG-225C	TA					
3056 (I-6)	CS-563E, CS-573E, CS-583E, CS-663E, AP-655C, CS-683E, CP-563E, CP-573E, CP-583E, CP-663E, M316C, M318C, M322C	ATAAC	DI	100 × 127	**3.94 × 5.0**	6.0	**365**
3066 (I-6)	320C, 320C L, 320C LN, 320C S, 321C LCR	T	DI	102 × 130	**4.0 × 5.1**	6.4	**391**

*Japan sourced.
**France sourced.
***Turbo optional on some models.
DI — Direct Injection
IDI — Indirect Injection
T — Turbocharged
TA — Turbocharged and Aftercooled
NA — Naturally Aspirated
ATAAC — Air/Air Aftercooled

NOTE: Materials and specifications subject to change without notice. Component commonality of Cat engines for all applications does not imply complete interchangeability. Contact your Cat dealer for specific information.

ENGINE/MACHINE CROSS REFERENCE (Continued)

Engine Model and (Cylinders)	Machine	Aspiration	Fuel Injection System	Bore × Stroke mm	Bore × Stroke in	Displacement L	Displacement in³
3116 (I-6)	CB-634D, BG-240C, AP-900B	T	DI	105 × 127	**4.13 × 5.0**	6.6	**402**
	BG-260C, BG-245C, AP-1050B, AP-1055B, 120H STD, 135H STD, BG-2455C, AP-1000B	TA	DI				
3126 (I-6)	D5*, D6N*, 561N, 953C, 963C	T	DI	110 × 127	**4.33 × 5.0**	7.2	**442**
	525B, 535B	TA					
	325C LN	ATAAC					
3304 (I-4)	527, 517	TA	DI	121 × 152	**4.75 × 6.0**	7.0	**425**
C3.4B	311F L, 313F L GC	T	DI	99 × 110	**3.90 × 4.33**	3.4	**207**
C4.4 ACERT	D3K, D4K, D5K, D6K2, 416F, 420F, 430F, 450F, 428F, 432F, 434F, 444F, 910K/M, 914K/M, 918M	T	DI	105 × 127	**4.13 × 5.0**	4.4	**268**
	318D2 L, 320D2 GC	TA					
	312E, 313D2/313D2 L, 313D2 GC, 313F L, 314E/314E L, 315F L, 316E L, 316F L, 318F L, 320F, 323F N, 325F	ATAAC					
C6.6	R1300G II	ATAAC	DI	105 × 127	**4.13 × 5.0**	6.6	**402**
C6.6 ACERT	D6N, 953D, 963D, 120M, PL61, M318D MH, M322D MH, 120M, 120M AWD, 501HD, 320E, 320E RR	TA	DI	105 × 127	**4.13 × 5.0**	6.6	**402**
C7 ACERT (I-6)	950H, 962H, IT62H, M325D MH, M325D LMH, 120K, 120K2, 12K, 12M, 140K, 140K2, 140M, 160K, 324D FM[2], 324D FM LL[2], 325D FM[2], 325D FM LL[2]	ATAAC	DI	110 × 127	**4.33 × 5.0**	7.2	**442**
C7.1 ACERT	320D Series 2 FM[2], 320D Series 2 FM LL[2], 538[1], 538 LL[1], 558 LL[1], 924K, 926M[1], 930K, 930M[1], 938K, 938M[1], 950 GC, 950K, 950L, 950M, 962K, 962L, 962M, MH3037[1], 120M2, 120M2 AWD, D5R2, D5T, 953K, 963K, 320D2, 323D2, 326D2, 330D2, 326F, 330F, 335F	ATAAC	DI	105 × 135	**4.1 × 5.3**	7.01	**427.8**
C7.2 ACERT	MH3037[2]	ATAAC	DI	110 × 127	**4.33 × 5.0**	7.2	**442**
C9 ACERT (I-6)	336D2, 336D2 XE, 340D2, 814F II, 815F II, D6T, MH3049, MH3059, 521B, 522B, 541 Series 2, 552 Series 2	ATAAC	DI	112 × 149	**4.4 × 5.9**	8.8	**537**
C9 ACERT (I-6)	627G Sc., 637G Sc., 973C, 140M AWD, 160M, 160M AWD, D6R2, D7R	TA	DI	112 × 149	**4.4 × 5.9**	8.8	**537**

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
[2] Emits equivalent to Tier 3/Stage IIIA/Japan 2006 (Tier 3).
*Not sold in U.S., Canada or Europe.
DI — Direct Injection
T — Turbocharged
TA — Turbocharged and Aftercooled
ATAAC — Air/Air Aftercooled

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ENGINE/MACHINE CROSS REFERENCE (Continued)

Engine Model and (Cylinders)	Machine	Aspiration	Fuel Injection System	Bore × Stroke mm	Bore × Stroke in	Displacement L	Displacement in³
3176 (I-6)	572R2, 345B L Series II, D7R, 814F, 815F, 816F	ATAAC	DI	125 × 140	**4.92 × 5.5**	10.2	**629**
C9.3 ACERT	D6T[1], D7E, 568[2], 568 LL[2], 966K, 966L, 966M, 966M XE, 972K, 972L, 972M, 972M XE, 336F, 336F XE, 12M2, 12M2 AWD, 12M3, 12M3 AWD, 140M2, 140M2 AWD, 140M3, 140M3 AWD, 160M2, 160M2 AWD, 160M3, 160M3 AWD, PL72	ATAAC	DI	115 × 149	**4.53 × 5.87**	9.3	**567.5**
3306 (I-6)	R1300G, 12H STD, 140H STD, 160H STD	T	DI	121 × 152	**4.75 × 6.0**	10.5	**638**
	D7G	TA	DI				
	545	ATAAC	DI				
3196 (I-6)	365B L Series II	ATAAC	DI	130 × 150	**5.1 × 5.9**	12.0	**732**
3406 (I-6)	RM-250C, RM-350B, D8R, D8R LGP	TA	DI	137 × 165	**5.4 × 6.5**	14.6	**893**
	826G Series II, 825G Series II, 824G Series II, AD30	ATAAC	DI				
C11 (I-6)	725, 730, 730 Ejector	ATAAC	DI	130 × 140	**5.1 × 5.5**	11.2	**680**
C11 ACERT (I-6)	R1600H, AD22, 966H	TA	DI	130 × 140	**5.12 × 5.51**	11.1	**680**
		ATAAC					
C13 ACERT (I-6)	R1700, 14M3, 16M3, 18M3, 349D2, 349F, 352F, 349F XE, 352F XE, 621K, 623K, 627K Tr, 972H, 980K, 980L, 980M, 982M	TA	DI	130 × 157	**5.12 × 6.18**	12.5	**763**
C15 ACERT (I-6)	D8T, D8T LGP, PL83, PL87, 657G Sc., 770G[1], 770G, 986H, 374F	TA	DI	137 × 172	**5.4 × 6.75**	15.2	**928**
	R2900G, R3000H, AD30, 735, 740, 740 Ejector, 824K, 825K, 826H, 980H	ATAAC	DI	137 × 171.5	**5.4 × 6.75**	15.2	**928**
3456 (I-6)	834G, 836G, 988G, 385B, 385B L, 5090B	ATAAC	DI	140 × 171	**5.5 × 6.75**	15.8	**966**
3408 (V-8)	D9R, 589, PM-565B	TA	DI	137 × 152	**5.4 × 6.0**	18.0	**1099**
		ATAAC					
C18 ACERT	AD45B, AD55, D9T, 631G, 637G Tr., 657G Tr., 988H, 988K, 772G[1], 772G, 834H, 834K, 836H, 385C MH, 24M, 390F	TA	DI	145 × 183	**5.7 × 7.2**	18.1	**1106**

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
[2] Emits equivalent to Tier 4 Interim/Stage IIIB/Japan 2011 (Tier 4 Interim).
DI — Direct Injection
T — Turbocharged
TA — Turbocharged and Aftercooled
ATAAC — Air/Air Aftercooled

NOTE: Materials and specifications subject to change without notice. Component commonality of Cat engines for all applications does not imply complete interchangeability. Contact your Cat dealer for specific information.

ENGINE/MACHINE CROSS REFERENCE (Continued)

Engine Model and (Cylinders)	Machine	Aspiration	Fuel Injection System	Bore × Stroke mm	Bore × Stroke in	Displacement L	Displacement in³
C27 ACERT (V-12)	AD60, D10T2, 773G[1], 773G, 775G[1], 775G, 990H, 990K, 844H, 844K, 6015B, 6030, 6030 FS	TA	DI	137 × 152	**5.4 × 6.0**	27.0	**1648**
3412 (V-12)	D10R, 5110B	TA	DI	137 × 152	**5.4 × 6.0**	27.0	**1649**
	844, 773E	ATAAC	DI				
3508 (V-8)	D11R, 5130B, 992G, 854G, 777D	TA	DI	170 × 190	**6.7 × 7.5**	34.5	**2105**
3512 (V-12)	785C, 6060, 6060 FS	TA	DI	170 × 190	**6.7 × 7.5**	51.8	**3158**
3512 HD (V-12)	785D	ATAAC	DI	170 × 215	**6.7 × 8.5**	58.6	**3574**
3516 (V-16)	789C, 994D, 5230B	TA	DI	170 × 190	**6.7 × 7.5**	51.8	**3158**
3516 HD (V-16)	789D, 994K	ATAAC	DI	170 × 215	**6.7 × 8.5**	78.1	**4765**
	793D, 994F, 994H	TA	DI	170 × 215	**6.7 × 8.5**	78.1	**4765**
C32 ACERT	777G[1], 777G, 854K, 992K, 993K, D11, D11 CD, 6020B, 6040, 6040 FS	TA	DI	145 × 162	**5.7 × 6.4**	32.1	**1959**
C175-16 (V-16)	793F, 794 AC, 795F	ATAAC	DI	175 × 220	**6.9 × 8.7**	84.7	**5167**
C175-16 (V-20)	797F	ATAAC	DI	175 × 220	**6.9 × 8.7**	105.8	**6458**

[1] Meets Tier 4 Final/Stage V/Japan 2014 (Tier 4 Final) emission standards.
DI — Direct Injection
TA — Turbocharged and Aftercooled
ATAAC — Air/Air Aftercooled

NOTE: Materials and specifications subject to change without notice. Component commonality of Cat engines for all applications does not imply complete interchangeability. Contact your Cat dealer for specific information.

TECHNOLOGY PRODUCTS

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INTRODUCTION

Caterpillar has long been a leader in bringing innovation to Cat products. Today, one of our biggest differentiators isn't just in our equipment — it's in the technologies we are integrating into our products that are improving machine performance and productivity, and transforming the way customers work to manage their operations and business more efficiently and cost effectively.

Construction Technology products deliver dramatic improvements on the job site to help customers manage their business more efficiently and cost effectively. Fully integrated into Cat machines and core systems, these technologies use advanced electronics to accurately measure and control productivity, monitor and manage machine health and maintenance, and provide more insight into your equipment fleet and operation.

Energy & Transportation Technology products simplify equipment management to help customers monitor and manage assets, operate at peak performance, and lower operating costs.

Cat Technologies combine the latest in technologies and dealer services to help customers improve in four key areas: Equipment Management, Productivity, Safety, and Sustainability to take optimum control of the job site. Whether running one machine, managing a fleet, or overseeing several job sites, Cat dealers can help customers with the right combination of technologies and services to meet the specific needs of their business and realize significant gains in productivity, efficiency and profitability.

- Cat Construction Technologies
  - Cat Grade
  - Cat Payload
  - Cat Compact
  - Cat Detect
  - Cat Link
  - Cat Command
- Cat Technologies
  - Equipment Management
  - Productivity
  - Safety
  - Sustainability

Mining Technology & Autonomy products continue to have a positive impact on mining operations around the world. Caterpillar offers a complete suite of technology products purpose built for the harsh mining environment. Our products combine the latest GNSS positioning technology with sophisticated electronic controls and software to help miners increase productivity, monitor fleet health and lower operating costs.

Cat MineStar provides the most comprehensive suite of mining technology products in the industry. It consists of a number of configurable capability sets that allow you to scale the system to your mine site needs. Cat MineStar helps manage everything from material tracking to sophisticated real-time fleet management, machine health systems, autonomous equipment systems and more. The capability sets — Fleet, Edge, Terrain, Detect, Health and Command — can be used in combination or individually to allow your operations the flexibility and scalability it needs to be more productive, efficient and safe.

MineStar Edge is a new platform that augments our current MineStar offerings and ensures we will be able to advance them for years to come. Edge delivers more connected, integrated, scalable and intuitive products that extend further up and down the value chain. Edge current offerings include Equipment Tracking and Production Recording.

- Cat MineStar
  - Fleet
  - Terrain
    - Terrain for Drilling
    - Terrain for Grading & Loading
  - Detect
    - Vision
    - Object Detection
    - Driver Safety System (DSS)
  - Health
    - Product Link Elite
    - Product Link Elite Data Streaming
    - VIMS Forwarder
    - VIMS Converter
    - VIMS Telemetry
    - VIMS Transmitter
    - Equipment Insights
    - Office
    - Technician Toolbox
    - PitLink
  - Command
    - Command for dozing
    - Command for underground
    - Command for hauling
    - Command for drilling
  - Edge
    - Equipment Tracking
    - Production Recording

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat Link Technology	**Cat Link** enables you to remotely monitor and manage your equipment so you can reduce costs, increase job site efficiency, and effectively manage your business.	All types of equipment.
Product Link™	**Product Link** wirelessly connects you to your equipment giving you valuable insight into how your machine or fleet is performing. The system communicates location hours, fuel usage, productivity, idle time, and diagnostics codes. Available with cellular and satellite connectivity.	Standard on many types of Cat equipment. Retrofits available on Cat or mixed fleet equipment.
VisionLink®	**VisionLink** is a telematics software application that enables you to remotely monitor your equipment so that you can make timely, fact-based decisions to maximize efficiency, improve productivity, and lower the cost of owning and operating your fleet.	Standard on many types of Cat equipment. Retrofits available on Cat or mixed fleet equipment.
Basic Health or Advanced Health (Formerly VIMS™)	Provides operators, maintenance, and fleet managers with vital machine health and production information.	Basic Health or Advanced Health is available on select Trucks, Track-Type Tractors, Wheel Loaders, Wheel Dozers, Motor Graders, and K Series 4-Drum Soil Compactors.
Cat Grade Technology	**Cat Grade** combines digital design data, in-cab guidance, and on some machines — automatic control — to help operators work more productively and accurately with less rework.	Select models of earthmoving, excavation, and paving equipment.
Cat AccuGrade™	**Cross Slope** systems are used for indicate-only guidance or to control one side of the blade to achieve accurate surface cross slope without any off-board infrastructure. This foundational system for motor graders can be combined with sonic, laser, GNSS or UTS technologies to make fine grading more efficient and productive. Machine mounted sensors are used to calculate necessary blade slope positioning. The system makes automatic adjustments to the left or right lift cylinder. The in-cab display delivers all of the cross slope information the operator needs to quickly and easily spread or cut material at the correct cross slope. The operator can select which side of the blade to control automatically and swap direction on the return pass without readjusting the settings. Elevation is controlled manually by matching grade or automatically by adding an elevation control device (sonic or laser). The cross slope systems are ideal for maintaining accurate cross slope over long distances without the need for infrastructure.	Motor Graders
	Global Navigation Satellite Systems (GNSS) use satellite technology to deliver precise 3D blade elevation and blade tip positioning information to the operator in the cab. Using machine-mounted components, an off-board GNSS base station and accuracy up to Real Time Kinematic (RTK) positioning, GNSS provides the information necessary for the system to accurately determine blade positioning with centimeter level accuracy. GNSS systems compute the positioning information on the machine compared to the position of the implement or machine relative to the design plane and delivers the information to the operator via an in-cab display. GNSS systems put all the information the operator needs to complete a job in the cab, resulting in a greater level of control. Provides precise location and elevation information for complex 3D contour applications, such as golf courses and highway super elevated curves.	Asphalt Compactors, Hydraulic Excavators, Motor Graders, Soil Compactors, Track-Type Tractors, Wheel Tractor-Scrapers, Landfill Compactors
	Laser systems provide precise 2D elevation control for grading with tight tolerances using a laser transmitter and receiver(s). A laser transmitter is set up on the work site to create a constant grade reference over the work area. A digital laser receiver is mounted on the blade of the machine and senses the laser signal as the machine moves across the work site. Laser systems are ideal for the fine grading on job sites with flat, single or dual slope surfaces, such as industrial, commercial or residential building sites.	Compact Track Loaders, Motor Graders, Multi Terrain Loaders, Skid Steer Loaders, Track-Type Tractors

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat AccuGrade	**Indicate-only** grade control systems for excavators provide in-cab guidance to indicate how much to cut or fill to achieve accurate grades and slopes. These systems can be combined with laser, GNSS or UTS technologies to make excavating more efficient and productive. Indicate-only systems provide the operator with precise real-time bucket positioning compared to an elevation reference to indicate how far the cutting edge is above or below desired grade. Using a combination of front linkage (stick, boom and bucket) sensors, the system calculates bucket tip position relative to a grade/survey stake or benchmark. When combined with a laser and receiver, the machine can travel after obtaining a reference benchmark from a laser transmitter to calculate the desired grade for elevation changes over large work sites, significantly increasing productivity. Indicate-only systems combined with laser are ideal for excavations, trenches, and general utility applications.	Hydraulic Excavators
	Sonic systems provide 2D "elevation" control using an ultrasonic sensor to maintain accurate blade height. The system traces a previous pass, string line, or curb and gutter and uses the elevation as a reference. The grade control system calculates blade adjustments based on the reference and desired elevation and automatically moves the blade to achieve grade. The operator simply steers the machine to maintain the sonic tracer over the external reference. Sonic systems are ideal for controlling elevation in fine grading applications where curb and gutter or stringlines are used as references such as road construction and road maintenance.	Motor Graders
	Sonic, Contact and Slope Sensors are combined to provide an elevation control system for paving and milling operations designed to control grade elevation and slope. The goal of grade and slope controls is to remove irregularities from the surface for maximum smoothness and control mat thickness for asphalt pavers and cutting depth for cold planers. The sonic system uses multiple transducers to provide an average of the reference surface while the contact sensor references the actual target to maintain the implement at the same relative vertical distance to an external reference such as a string line or a curb and gutter or even existing pavement. Slope Sensors complement these systems and are designed to control cross slope of the screed or cold planer. These sensors can work individually or together and make automatic elevation adjustments to maintain the appropriate targeted values. The operator simply steers the machine to maintain the sensor over the external reference and monitors the remaining job site variables to ensure a consistent product. Referencing beams are also available both as contact as well as non-contacting styles which help to provide better averaging of the existing surface to help customers achieve their smoothness targets.	Asphalt Pavers, Cold Planers
	Universal Total Station (UTS) systems use high-accuracy, dynamic tracking technology on the job site to track a target mounted on the implement of the machine to determine the most precise 3D positioning. The system uses active target technology to reliably lock onto and track the intended target, eliminating false lock-ons to other active machine targets, survey crews or reflective surfaces. The UTS instrument continuously measures the target's position and transmits real-time positioning data to the operator via the in-cab display, which shows the exact position of the implement in relation to the design. The system combines the position of the target with the known position of the implement, blade or tips, machine measurements and sensor outputs to calculate precise positioning of the blade tips. The system uses the positioning data to calculate desired elevation and cross slope. Cut and fill values are computed by comparing the positioning of the blade with the design file.	Asphalt Pavers, Hydraulic Excavators, Motor Graders, Track-Type Tractors, Cold Planers

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat Grade with Cross Slope	**Cross Slope** systems are used for indicate-only guidance or to control one side of the blade to achieve accurate surface cross slope without any off-board infrastructure. This foundational system for motor graders can be combined with sonic, laser, GNSS or UTS technologies to make fine grading more efficient and productive. Machine mounted sensors are used to calculate necessary blade slope positioning to achieve desired cross slope of the surface. The system makes automatic adjustments to the left or right lift cylinder, typically performed by the operator. Cat Grade Control Cross Slope systems are factory integrated and utilize the existing, standard machine display reducing the need for an additional grade control display for cross slope guidance. The in-cab display delivers all of the cross slope information the operator needs to quickly and easily spread or cut material at the correct cross slope. The operator can select which side of the blade to control automatically and swap direction on the return pass without readjusting the settings. Elevation is controlled manually by matching grade or automatically by adding an elevation control device (sonic or laser). The cross slope systems are ideal for maintaining accurate cross slope over long distances without the need for infrastructure.	Motor Graders
Cat Grade with 3D	**3D** systems use Global Navigation Satellite System (GNSS) satellite technology to deliver precise blade positioning and location information to the operator in the cab. Using machine-mounted components, position sensing hydraulic cylinders, an off-board GNSS base station, and Real Time Kinematic (RTK) positioning accuracy, these systems provide the information necessary to determine precise blade positioning with centimeter level accuracy. The system computes the positioning information on the machine compared to the position of the cutting edge relative to the design plane and delivers the information to the operator via the standard in-cab display. All the information the operator needs to complete a job is delivered to the cab display, resulting in a greater level of control. Cat Grade Control 3D systems also provide overcut protection and linkage to additional Caterpillar exclusive features like AutoCarry™, Automatic Ripper Control and Load Assist. These systems are ideal for high production dozing and cutting complex 3D contours, such as highway projects.	Track-Type Tractors, Wheel Tractor-Scrapers
Cat GRADE with Depth and Slope	The **2D indicate-only** system for excavators provides the operator with precise real-time bucket positioning relative to the desired grade. Using a combination of position sensing hydraulic cylinders and sensors on the stick and boom pins, the system calculates bucket tip position relative to a grade/survey stake or benchmark. When used with the included laser receiver, the machine can reference a benchmark and travel while maintaining the elevation reference, even on uneven terrain. The laser receiver references the laser from a transmitter on the job site and calculates the desired grade for elevation changes, significantly increasing productivity. The factory-integrated system uses the existing machine display, reducing the need for an additional display. Sensors are deeply integrated and protected from damage in rugged applications. Cat Grade Control Depth and Slope is ideal for maintaining accurate grades and slopes in excavations and trenching applications.	Hydraulic Excavators

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat Grade with Slope Assist	**Slope Assist** systems provide blade slope/angle control to the cutting edge to achieve accurate slopes with a quality surface finish in less time with less effort — without the need for off-board infrastructure. This integrated machine system uses a blade mounted sensor to calculate necessary blade slope position to achieve desired surface slope and main-fall. The system makes automated adjustments to the lift and tilt cylinders, typically performed by the operator. The in-cab display uses the existing machine display to deliver the information the operator needs to quickly and easily spread or cut material at the correct angle. Two modes are available: basic and advanced. Basic mode maintains the slope of the last given blade command. Advanced mode drives to and maintains a preset target slope. Elevation is controlled manually by matching grade or automatically by adding an elevation control device. This foundational system for dozers can be combined with laser, GNSS or UTS technologies to make grading even more efficient and productive.	Select Track-Type Tractors
Cat Grade and Slope	The **Grade and Slope** system for asphalt pavers and cold planers utilize a combination of sonic, contact and slope sensors to provide an elevation control system for paving and milling operations designed to control grade elevation and slope. The goal of grade and slope controls is to remove irregularities from the surface for maximum smoothness and control mat thickness for asphalt pavers and cutting depth for cold planers. The sonic system uses multiple transducers to provide an average of the reference surface while the contact sensor references the actual target to maintain the implement at the same relative vertical distance to an external reference such as a string line or a curb and gutter or even existing pavement. Slope Sensors complement these systems and are designed to control cross slope of the screed or cold planer. These sensors can work individually or together and make automatic elevation adjustments to maintain the appropriate targeted values. The operator simply steers the machine to maintain the sensor over the external reference and monitors the remaining job site variables to ensure a consistent product. Referencing beams are also available both as contact as well as non-contacting styles which help to provide better averaging of the existing surface to help customers achieve their smoothness targets. Cat Grade and Slope for asphalt pavers includes vandal protection of the display boxes. Both asphalt pavers and cold planers come factory calibrated, utilize robust harnesses with strain relief components for excellent reliability and durability — a crucial element for paving applications.	Asphalt Paver, Cold Planers
Cat Compact Technology	**Cat Compact** combines advanced compaction measurement, in-cab guidance and reporting capabilities to help consistently meet compaction targets faster, more uniformly, and in fewer passes — reducing rework and material costs in both soil and asphalt applications.	Available on select Cat compactors.

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat Compact	Compaction measurement is the foundation system for vibratory soil compactors. Compaction measurement is a real time indication of soil stiffness, provided by one of two available technologies: accelerometer based Compaction Meter Value (CMV) or Caterpillar exclusive energy based Machine Drive Power (MDP). The compaction measurement system outputs to the standard multipurpose display, providing the operator with real time data that helps them determine the state of compaction. The scalable compaction measurement system can be upgraded to include mapping. Mapping from the factory is an integrated SBAS accuracy GNSS which uses a satellite technology to deliver drum positioning information to the operator in the cab. The factory integrated design protects expensive and fragile components of the system from damage, theft or vandalism in a way that aftermarket systems cannot. The SBAS system from the factory does not require base stations or other off board hardware, but can be easily upgraded in the field with an AccuGrade RTK radio for greater accuracy. Mapping capability allows the operator to record and map compaction measurement values and pass counts, as well as other compaction application data for future analysis. Cat Compaction Control for asphalt compactors and the CW34 pneumatic tire roller provides mat temperature mapping and pass-count information to the operator, ensuring that compaction occurs at the optimum mat temperature and that uniform coverage is completed.	B Series Vibratory Soil Compactors, K Series 4-Drum Soil Compactors, A and B Series Asphalt Compactors, CW34 Pneumatic Tire Roller
Payload Technology	**Cat Payload** enables accurate weight measurement of material being loaded or hauled. Payload information is shared with loader operators in real-time to improve productivity, reduce overloading and record both weights and number of loads per shift.	Available on select Cat wheel loaders, trucks, excavators and scrapers.
Cat Payload	**Cat Payload** provides on-the-go payload weighing through an in-cab LCD display so operators can deliver exact loads with confidence and work more efficiently. Operators can fill trucks to capacity without overloading, increasing productivity and eliminating costly fines. The system records detailed data to track productivity, including time/date, weights, cycles, and more. An optional in-cab printer enables operators to print individual load tickets as well as truck and material summary reports. Daily summary reports in the *VisionLink* web portal and detailed payload reports in *VIMSpc* software are available for productivity reporting and analysis.	Wheel Loaders
Cat Payload with Production Measurement	**Production Measurement** brings on-the-go payload weighing to cab to help operators deliver exact loads with confidence and work more productively and accurately; without rework. The system is integrated at the factory into the standard cab display. Operators can track load weights in real time and know precisely how much material is in the bucket or truck. Instant payload feedback gives operators the confidence to know when loads are filled to target capacity, which reduces under and overloading, and maximizes the potential of the entire fleet. Payload information is stored in the display, enabling operators to track productivity such as weights, and totals per shift while in the cab. Productivity data can be accessed wirelessly through the VisionLink web portal giving the site supervisor a daily view of production totals and efficiency metrics.	Wheel Loaders, Articulated Trucks, Excavators

PRODUCT	CONSTRUCTION TECHNOLOGY APPLICATIONS	EQUIPMENT
Cat Payload with Estimator	**Payload Estimator** uses bowl lift cylinder pressure on the loaded haul segment to calculate payload. This integrated system provides on-the-go payload weighing through the standard machine display so operators can deliver exact loads with confidence and work more efficiently. Deep integration ensures reliable operation and accurate data (within ±5%). It is optimized to work with Sequence Assist or can be used in manual mode. Productivity data can be accessed wirelessly through VisionLink, giving the site supervisor a daily view of production totals.	Wheel Tractor-Scrapers
Cat Payload with Truck Production Management System (TPMS)	**Truck Production Management System (TPMS)** enhances truck and loader effectiveness for improved fleet productivity and reduced operating cost. TPMS offers external lights or an optional digital display to signal the loading tool operator when the proper load is reached. The on-board system provides a payload accuracy of ±5% and stores payload weight, cycle segment times, cycle segment distance and fuel, operator ID and actual clock time and date of each cycle.	Off-Highway Trucks
Cat Detect Technology	**Cat Detect** combines safety and monitoring systems to enhance operator awareness. By expanding your view of the working environment around your equipment, you can improve the safety and productivity of your entire operation and keep your people and assets safe.	Available on many models of Cat equipment.
Cat Detect with Rear Vision Camera	**Rear vision cameras** greatly enhance visibility behind the machine to help the operator work more productively. On many machines, the camera view is displayed through the standard display.	Standard on many types of Cat equipment. Retrofits available on most models.
Cat Detect with Work Area Vision System (WAVS)	**Work Area Vision System (WAVS)** is used on larger equipment in applications where there are multiple machines, such as in a quarry. WAVS uses up to three cameras to provide full coverage in areas of limited visibility. Views can be set up by camera location (front, rear, side) or programmed to switch automatically based on direction of travel.	Optional on larger trucks and wheel loaders.
Cat Detect with Tire Monitoring	**Tire Monitoring** uses sensors located on the rims to display tire pressures and temperatures on the in-cab monitor to help keep fleets running safely and productively. Operators can view real-time information via the Messenger display and work with confidence. Alerts enable operators to take immediate action before a tire failure occurs to reduce downtime. The system optimizes fuel consumption by operating at optimum tire pressures and reduces tire damage and frequent replacements due to high tire temperatures and underinflation. Caterpillar designed and tested aftermarket kits work with multiple machine models and mixed fleets. Remote monitoring via VisionLink helps equipment managers schedule maintenance and make informed decisions.	All Cat and mixed fleet machines with tubeless tire rim sizes R29.5 and smaller.
Cat Detect with Machine Security System (MSS)	**Machine Security System (MSS)** prevents unauthorized personnel from starting the machine by using a uniquely coded key that is programmed to a computer chip in the MSS module on-board the machine. The system can be set to prevent unauthorized use outside of normal working hours, and protect from vandalism or theft.	Standard on many types of Cat equipment.

PRODUCT	ENERGY & TRANSPORTATION TECHNOLOGY APPLICATIONS	EQUIPMENT
	Energy & Transportation	
Product Link Web	Product Link Web provides features optimized for power systems equipment, including detailed engine parameters. An innovative tool for equipment management, the Product Link Web user-friendly interface enables communication between the customer and their equipment assets, and provides comprehensive information about the performance and condition of the equipment. It also enables the user to customize alerts so that critical operations can be prevented and or receive immediate attention.	Oil and Gas, Marine, Electric Power Generation, and Industrial installations
Cat AssetIQ™ Kits	AssetIQ Kits are field-installed solutions for collecting engine data and information on Cat or other manufacturers' mechanical or basic electronic equipment. Kits are available for Fuel Flow Monitoring Systems and Sensor Kits for basic engines and genset applications.	Oil and Gas, Marine, Electric Power Generation, and Industrial installations

PRODUCT	MINING TECHNOLOGY & AUTONOMY APPLICATIONS	EQUIPMENT
	Cat MineStar	
Fleet	Fleet is a comprehensive mine management system featuring an advanced truck assignment engine, health and operational event alarming, "what-if" analysis, productivity tracking, machine tracking, material management and a comprehensive reporting package. Integrating with other Cat MineStar capability sets, Fleet improves productivity by 10-15%, eliminates misdirected loads, improves information availability and provides greater flexibility to adjust for changing mine and market conditions.	Entire mining fleet
Terrain	Terrain for drilling increases hole placement and depth accuracy while removing the cost of drill pattern survey and staking. Position and status information of other drills working on the same pattern is provided to operators in real time on the in-cab display. Terrain for grading ensures accurate execution of the design plan and enables safe operating practices. It can be used on a variety of machines in numerous applications from production dozing to reclamation, all helping you mine more safely and productively. Terrain for loading provides grade indication for bench management and automatic material information on each pass helping operators to move the right amount of material with every bucket load. Safety, productivity and efficiency are positively impacted and the mine design plan accurately executed.	Blasthole Drills, Articulated Drills, Draglines, Scrapers, Loaders, Dozers, Shovels, Motor Graders, Hydraulic Excavators, Track-Type Tractors, Surface Miners, Terrain Levelers

PRODUCT	MINING TECHNOLOGY & AUTONOMY APPLICATIONS	EQUIPMENT
	Cat MineStar	
Detect	Vision is a camera system available on mobile surface equipment. It provides multiple camera views that improve the operator's awareness of the vicinity around working equipment.	Surface: Select models of Electric Rope Shovels, Hydraulic Shovels, Off-Highway Trucks, Medium Wheel Loaders, Large Wheel Loaders, Large Motor Graders, Medium Wheel Dozers, Large Wheel Dozers
	Object Detection uses cameras and radars to improve the operator's view of key areas around the machine, allowing operators to make informed decisions. The system works with several machine types increasing machine perimeter awareness during critical periods including startup, initial movement and reverse travel. The system consists of an interactive touch screen display, radars and cameras on the front, rear and sides of the machine. Radar and camera configuration will vary by machine type.	Select models of Off-Highway Trucks, Medium Wheel Loaders, Large Wheel Loaders, Large Motor Graders, Medium Wheel Dozers, Large Wheel Dozers
	The Driver Safety System (DSS) is a non-intrusive, in-cab fatigue detection technology that instantly alerts operators the moment fatigue or distraction is identified. Fatigue detection technology works by monitoring eye-closure duration and head pose. If the DSS detects a fatigue or distraction event the operator is immediately alerted through configurable in-vehicle seat vibration and/or audio alarm. The fatigue or distraction event is sent to a 24-hour monitoring center to classify and confirm the event. Caterpillar experts will analyze the data and provide customized reporting with site-level recommendations. By cross-referencing fatigue and distraction events against available equipment data, Caterpillar can provide suggestions to improve operational efficiency.	Surface and underground mining machines as well as On-Highway Trucks
Health	Health delivers critical event-based equipment condition and operating data for your entire fleet. It includes comprehensive, proactive health and asset monitoring capabilities, with a wide range of diagnostic, reporting tools, analytics and recommendations.	Cat mining equipment and select models of other brands of mining machines

PRODUCT	MINING TECHNOLOGY & AUTONOMY APPLICATIONS	EQUIPMENT
	Cat MineStar	
Command	Command for hauling enables total autonomous operation of large mining trucks. Advanced perception and sensing technologies enable self-driving trucks within the autonomous operating zone to work safely and productively alongside manually operated equipment, such as loading tools, cleanup and maintenance machines, and mine site personnel. Utilizing a virtual mine map and dynamic routes managed from a central command center, autonomous trucks respond to calls to the shovel, move into loading position, haul loads to the designated dump points, and even report to the maintenance bay, all without an operator on-board.	Select models of Large Mining Trucks
	Operator Machine Assist (OMA) automates the entire drill cycle for a single-row, including tramming between holes, while the operator remains in the cab of the machine. OMA delivers higher consistent productivity than a typical staffed operation. Semi-Autonomous Drilling System (S-ADS) autonomously drills an entire row without an operator in the cab monitoring from a remote operations center or from a tablet interface. Flexible and simple, the system makes drilling autonomy easy to implement and more accessible for all kinds of operations. Command for drilling takes a building block approach to autonomy allowing customers to enter at their own pace.	Caterpillar Diesel MD6250 and MD6310 Caterpillar Electric MD6640 (Bucyrus 49R/HR) and Komatsu Electric P&H 120A and P&H 320XPC
	Command for underground is designed to enhance safety and boost operator efficiency and effectiveness. The system allows the operator to work from a safe and ergonomic work station far from the Load Haul Dump (LHD) machine — either on the surface or underground without sacrificing machine productivity. The system can also have a significant impact through increased machine availability, decreased operating costs and extended machine life.	Available as an attachment for select models of Hard Rock Underground Mining Loaders
	Command for dozing enables remote operation from a safe location away from the machine. Removing operators from the cab of a machine working in hazardous conditions promotes safety and reduces operator exposure to dust, noise and vibration. Command for dozing can utilize either an over-the-shoulder operator console or a seated operator station for longer term operations. Either system utilizes line of sight communications for near machine operation. Or with the addition of an on board vision system, the operator station can be located anywhere on-site or from an off-site command center. Terrain's avoidance zone functionality (optional) can be utilized to further enhance safety. Semi-Autonomous Operation allows an operator to control up to four machines from a Remote Operator Station while semi-automating a push-to-edge or pivot-push (late 2018) application.	D8T, D10T, D11, D11 CD Track-Type Tractors
Edge Equipment Tracking	Equipment Tracking is a fleet management system that replaces pencil and paper with automatic tracking of all mobile assets. It provides highly accurate data that tells supervisors what the fleet is doing, who is operating which machine and how they are spending their time.	Entire Mining Fleet
Edge Production Recording	Production Recording pairs with Equipment Tracking to deliver an accurate and automated near-real-time solution that measures and reports on every aspect of the load-haul-dump cycle. It gives sites visibility to their entire mining operation and provides accurate, reliable and actionable data with no operator input required.	Loading and Hauling Mining Fleet